

Blind Transmitter Localization in Wireless Sensor Networks: A Deep Learning Approach

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Abstract—This paper describes a blind transmitter localization technique based on the deep neural network (DNN) framework. Blind localization assumes no previous knowledge on the transmit signal. It is shown that DNN based location approaches the maximum likelihood solution with reduced computational complexity. Moreover, the maximum likelihood, least squares and radio environment map localization estimators are presented in order to compare the design and performance of the proposed DNN algorithm. The system model is built based on a wireless sensor network that collects received signal strength measurements assuming disturbances of distance dependent correlated shadowing noise. Performance evaluation using numerical simulations shows that the proposed DNN scheme achieves location accuracy similar to the optimum maximum likelihood estimator while presenting computational complexity reduction of more than 90%.

Index Terms—Blind localization, wireless sensor networks, received signal strength, deep learning, spectrum sensing.

I. INTRODUCTION

TRANSMITTER location information available to network orchestration entities represents an important step towards wireless networks that are not only able to better allocate time-frequency resources to users, but that also have the ability to make use of interference protection mechanisms. The need to build efficient, and yet powerful, wireless networks is a common objective for the recent advances in mobile communication standards. The use of shared resources is a key feature to achieve these goals. For instance, perspectives on beyond 5G research point that pencil sharp beams are likely to be available, and network side localization with centimeter level precision is crucial for fast beam alignment algorithms [1]. Therefore, location awareness paves the way towards proactive flexible radio resource management by the network. Moreover, private or enclosed wireless networks employed within factory plants or large farming areas, are particularly threatened by intentional or accidental jamming transmissions. Arguably, security cameras can already address the problem of localizing intentional jamming units. However, a trained security crew must be always vigilant to spot violations. Furthermore, privacy concerns are growing against cameras, while radio sensing is considered an alternative that is more suitable to preserve people's privacy. To this end, the physical transmitter position is a valuable information for allowing better distribution of resources among users, as well as to

detect and act against jamming from unauthorized transmitters [2], [3].

The classical localization techniques can be divided into ranging-based, angle-based and fingerprinting matching [4]. Ranging-based techniques use the estimated distances between transmitter and sensing units (SUs) for calculating the corresponding transmitter position via geometric multilateration, whereas angle-based techniques use angles for the same purpose. For the case of ranging-based localization, distance estimation can be obtained in two ways: via its relation with received signal strength (RSS), or with the wave propagation delay. The best mathematical models for RSS are stochastic, while for propagation delay the relation between distance and time of arrival (ToA) is a mere multiplication by the speed of light. Therefore, localization based on ToA has been regarded as more accurate when compared to RSS in most circumstances [5]. However, ToA estimation requires the transmission of a wide-band known sequence, and hence blind transmitter localization becomes impossible. For the angle-based, each SU must have an antenna array for estimating the angle of arrival (AoA), and the angular resolution depends upon the number of elements in the array, which increases SUs' size, cost and complexity. Consequently, RSS is particularly interesting for simple and low cost blind transmitter localization applications. Nevertheless, the performance of schemes that employ multilateration based on RSS distance estimation is particularly affected by random propagation effects, shadowing noise and hardware nonlinearities [6]. For this reason, localization schemes which are based on measurement database search, i.e., fingerprinting matching, have been proposed as alternatives [7]. However, fingerprinting implementations require an extensive search over the database, which requires significant computational resources from the central unit (CU). Given these points, the learning paradigm of deep neural networks (DNNs) seems to fit very well as a solution to the blind localization problem, since all the database information is compiled into a module that requires significantly less processing resources. Hence, this paper aims at evaluating the localization performance using DNN, and compares it against the well-known least squares (LS), maximum likelihood estimator (MLE) and the recently proposed radio environment map localization (REML). It is important

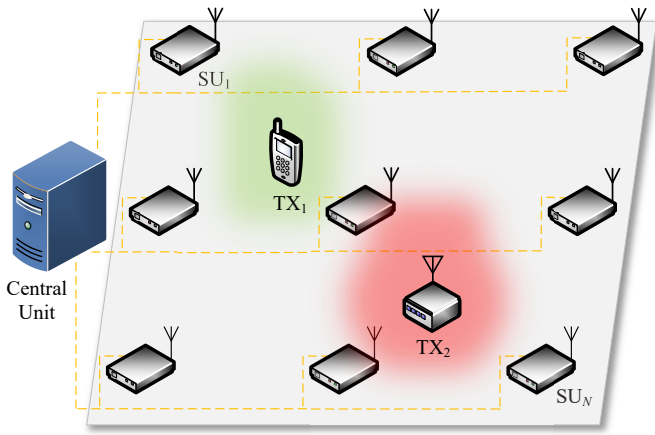


Fig. 1: System model illustration.

to point out that in order to make use of LS and MLE shown here, one needs to assume that the measurements are accurately described by a certain path loss model. However, such assumption might not hold entirely true, since it is quite optimistic to assume that a model can capture all the effects present in practical conditions. Therefore, DNN localization schemes that are based on a measurement database can play a significant role in boosting the localization performance, particularly in indoor environments. Furthermore, no assumption needs to be made with respect to the transmit power, thus making DNN blind localization feasible in real world applications.

The remainder of the paper is organized as follows: Section II presents the system model. Section III describes classical localization algorithms together with the DNN-based localization scheme. Section IV analyses the performance of the localization schemes. Finally, the paper is concluded in Section V with future insights, and directions on the topic of machine learning based localization algorithms.

II. SYSTEM MODEL

In the system model illustrated by Fig. 1, several SUs with known positions measure the RSS in a band of interest for a sufficient time interval to average out the small scale fading effects. It is assumed that the RSS at the j -th SU obeys the log-normal path loss model [8]. Let $\mathbf{m} = [P_1, \dots, P_N]^T$ represent the measurement vector, which contains the RSS measured by N SUs, where the j -th SU measurement can be expressed as

$$P_j = P_0 - 10\beta \log_{10} \left(\frac{d(\mathbf{u}, \mathbf{v}_j)}{d_0} \right) + n_j, \quad (1)$$

where P_0 (dBW) is the received power at a reference distance d_0 , β the environment dependent path loss exponent, n_j represents entries from a zero-mean Gaussian random vector $\mathbf{n}$ with covariance matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$, $\mathbf{u} \triangleq [u_x, u_y]^T$ contains the unknown transmitter coordinates in two dimensional space,

$\mathbf{v}_j \triangleq [v_{jx}, v_{jy}]^T$ the known coordinates of the j -th SU, and $d(\mathbf{u}, \mathbf{v}_j)$ represents the Euclidean distance, defined as

$$d(\mathbf{u}, \mathbf{v}_j) = \sqrt{(u_x - v_{jx})^2 + (u_y - v_{jy})^2}. \quad (2)$$

It is worth noting that the shadowing noise captures the effects of random movement and signal blockage in the environment, and therefore it depends on the position of the SUs. As a result, the correlation between shadowing noise n_a , and n_b depends on the distance between the a -th and b -th SU [9], and thus the covariance matrix entries can be expressed as

$$[\mathbf{C}]_{a,b} = \sigma_{\text{dB}}^2 \exp \left(-\frac{d(\mathbf{v}_a, \mathbf{v}_b)}{d_{\text{cor}}} \right), \quad (3)$$

where d_{cor} represents the decorrelation distance, and σ_{dB}^2 the shadowing noise variance.

The CU collects the measurements from all SUs, and this information is used for estimating the coordinates of one or more transmitters present in the area of interest. In order to evaluate the performance of different localization schemes, it is assumed that the links among SUs and the CU introduce negligible error to RSS measurements, and there is a single transmitter present in the area of interest. The main objective is to estimate as precisely as possible the coordinates of the transmitter present in the area of interest given the vector of power measurements obtained by N SUs.

III. TRANSMITTER LOCALIZATION TECHNIQUES

This section describes the classical MLE, LS and REML localization methods, and presents the proposed DNN-based scheme.

A. Maximum Likelihood Estimation

The MLE is optimal in the sense that it is asymptotically efficient, i.e., it becomes unbiased and attains the Cramér-Rao lower bound (CRLB) for sufficiently large observation data [10]. Without loss of generality, it is assumed the reference distance is one meter, and therefore given the system model described by (1), the MLE is obtained as

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg \min_{\boldsymbol{\theta}_{\text{ML}} \in \mathbb{S}} (\mathbf{m} - \mathbf{f}(\boldsymbol{\theta}_{\text{ML}}))^T \mathbf{C}^{-1} (\mathbf{m} - \mathbf{f}(\boldsymbol{\theta}_{\text{ML}})), \quad (4)$$

where $\boldsymbol{\theta}_{\text{ML}} \triangleq [\mathbf{u}, P_0, \beta]^T$ represents the unknown parameter vector, $\mathbb{S}$ the set of possible values each parameter can assume, and the j -th entry of $\mathbf{f}(\boldsymbol{\theta}_{\text{ML}})$ is given by

$$[\mathbf{f}(\boldsymbol{\theta}_{\text{ML}})]_j = P_0 - 10\beta \log_{10} (d(\mathbf{u}, \mathbf{v}_j)). \quad (5)$$

Even considering certain simplifications, such as prior knowledge on β and P_0 , the MLE does not have a closed-form solution. Therefore, suboptimal approaches rely on extensive grid search [2], iterative maximization of the likelihood function [11], or relaxation techniques [12].

B. Least Squares

The LS estimator for the transmitter coordinates is obtained by minimizing of the squared difference between the measured RSS collected by the SUs and a given system model for the measurements [11]. Unlike MLE, in the LS formulation there is no assumption about the probabilistic characterization of the data. Hence, this estimator is equally valid regardless the shadowing noise variance accurately follows the correlated Gaussian distribution. For deriving the estimator, let us consider $d_0 = 1$ m, and by dropping the shadowing noise term, (1) can be rewritten as

$$d^2(\mathbf{u}, \mathbf{v}_j) = \alpha z_j, \quad (6)$$

where $\alpha = 10^{P_0/5\beta}$ and $z_j = 10^{-P_j/5\beta}$. This can be further expressed in a matrix form as

$$\mathbf{X}\boldsymbol{\theta}_{\text{LS}} = \mathbf{y}, \quad (7)$$

where $\boldsymbol{\theta}_{\text{LS}} \triangleq [\mathbf{u}^T, \mathbf{u}^T \mathbf{u}, \alpha]^T$ represents the unknown parameter vector, and the entries of $\mathbf{X}$ and $\mathbf{y}$ are given by

$$\mathbf{X} = \begin{bmatrix} 2\mathbf{v}_1^T & -1 & z_1 \\ \vdots & \vdots & \vdots \\ 2\mathbf{v}_N^T & -1 & z_N \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_N^T \mathbf{v}_N \end{bmatrix}. \quad (8)$$

As a result, the LS estimator is given by

$$\hat{\boldsymbol{\theta}}_{\text{LS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (9)$$

C. Radio Environment Map Localization

In this scheme, the area to be monitored is divided into discrete points, or pixels, with a certain resolution R pixel/meter. The total number of pixels along the width and height of the area of interest are $P_w = A_w/R$ and $P_h = A_h/R$ pixels, where A_w and A_h represent the width and height in meters, respectively. The total number of pixels within the area is $K = (P_w \times P_h)$ pixels. For creating the REM, the RSS measurements are available for only the pixels that contain the SUs, these are used to estimate the RSS values in the remaining pixels. This estimation is carried out via ordinary Kriging interpolation [2]. The resulting REM is stored in a matrix $\mathbf{U} \in \mathbb{R}^{P_w \times P_h}$. The estimated transmitter pixel position is obtained by selecting the element which contains the highest RSS value as

$$\hat{\mathbf{u}}_{\text{pixel}} = \arg \max_{i,j} [\mathbf{U}]_{i,j}, \quad (10)$$

where the continuous space coordinates are obtained by $\hat{\mathbf{u}}_{\text{REML}} = \hat{\mathbf{u}}_{\text{pixel}}/R$. It is important to note that REML doesn't require knowledge of the transmit power, making it also practical for blind localization.

D. Deep Learning Based Transmitter Localization

1) *Deep Neural Networks*: Neural networks consist of a layered structure of linear and non-linear mathematical operations that present the potential to approximate an arbitrary function [13]. These operations are represented in a graph by layered non-linear units (neurons), weights associated with

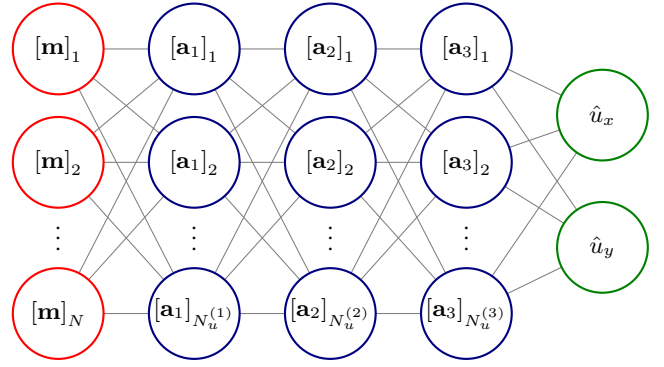


Fig. 2: Selected DNN architecture.

edges, and input and output units. A layer between the input and output is termed hidden layer. DNNs have more than one hidden layer, and this structure often yields a good trade-off between generalization capability and over-fitting on the training examples, i.e., high variance.

The training of DNN architectures is a topic of active research, and it has developed enormously in the recent years due to its success in tasks such as speech and image recognition. In short, the training process performs an iterative minimizing of a predefined loss function between estimated and known training examples, i.e., supervised learning. The mean squared error (MSE) is a natural measure between the estimated and true values in regression problems. However, the justification for using the MSE as a loss function is that, it is proved to yield to the MLE [13]. Therefore, the learned weights of the DNN will be an approximation to the MLE, but with very low online complexity. However, this comes at the price of an additional offline step, namely the *training* of the network.

It is important to note that, estimators obtained via data driven approaches are naturally immune to system modeling simplifications and misconceptions, since they are obtained readily from measured data sets or extended data sets. However, in this paper, the model of (1) is used to train and validate the performance of DNN estimators with numerical simulations.

2) DNN Architecture for Transmitter Localization:

Ranging-based localization methods first estimate the distances between the transmitter and the SUs, and use this information to obtain the location coordinates geometrically. However, this solution performs two steps in which errors can occur and degrade the estimation performance. Therefore, in the proposed architecture, the input layer uses the measurement vector directly.

Figure 2 illustrates the selected architecture, where the activation of the l -th hidden layer is given by

$$\mathbf{a}_l = \text{ReLU}(\mathbf{W}_l^T \mathbf{a}_{l-1} + \mathbf{b}_l). \quad (11)$$

Here $\mathbf{W}_l \in \mathbb{R}^{N_u^{(l-1)} \times N_u^{(l)}}$ and $\mathbf{b}_l \in \mathbb{R}^{N_u^{(l)} \times 1}$ contain the weights and biases associated with the edges connecting the

TABLE I: DNN hyperparameters

Hyperparameter	Value
Number of hidden layers	3
Number of hidden units per layer	128
Number of training examples	5000
Validation split	80% training, 20% validation
Mini batch size	32
Regularization parameter	0.01
Activation function of hidden units	Rectified linear unit (ReLU)
Epochs	2000
Optimizer	Adaptive moments (Adam) [15]
Learning rate	10^{-4}
Loss function	MSE (13)
Metric function	MPE (14)
Weight initialization	Xavier [16]

$(l - 1)$ -th to the l -th layer, respectively, $N_u^{(l)}$ is the number of units in the l -th layer, and $\text{ReLU}(x) = \max(0, x)$ is the rectified linear unit, which is the non-linear function associated with the hidden units. The last layer outputs the estimated transmitter coordinates with no further processing. Therefore, the estimated transmitter coordinates are given by

$$\hat{\mathbf{u}}_{\text{DNN}} = \mathbf{W}_L^T \mathbf{a}_{L-1} + \mathbf{b}_L. \quad (12)$$

The Python based application programming interface Keras has been employed for designing the architecture, training and testing. The MSE between the estimated and true transmitter coordinates as implemented in Keras [14] can be written as

$$\text{MSE} = \frac{1}{2m} \sum_{k=1}^m \sum_{i=1}^2 \left([\hat{\mathbf{u}}]_i^{(k)} - [\mathbf{u}]_i^{(k)} \right)^2 + \frac{\lambda}{2m} \sum_{w \in \mathbb{W}} w^2, \quad (13)$$

where m is the total amount of training examples in one training batch, λ is the regularization parameter and $\mathbb{W}$ represents the set which contains all weights from the network.

After intensive experiments, an architecture with 128 units in 3 hidden layers shows good generalization performance while keeping a moderate level of complexity for the scenarios considered, and no significant dependency on the number of sensors was observed. Table I presents all the hyperparameters used, which were selected during the tuning phase. In order to halt the training process, the mean position error (MPE) between the estimated and exact transmitters' coordinates is used as a metric, and it is defined as

$$\text{MPE} = \frac{1}{m} \sum_{k=1}^m d \left(\hat{\mathbf{u}}^{(k)}, \mathbf{u}^{(k)} \right). \quad (14)$$

The weights selected after a number of training epochs are the ones which have smallest MPE for both training and validation sets. Although the training process is the iterative minimization of the MSE, the weights and biases that present the smallest MPE are of practical interest. Note that the set of weights that yield the smallest MSE might not be the set that also yields the smallest MPE. Hence, with the

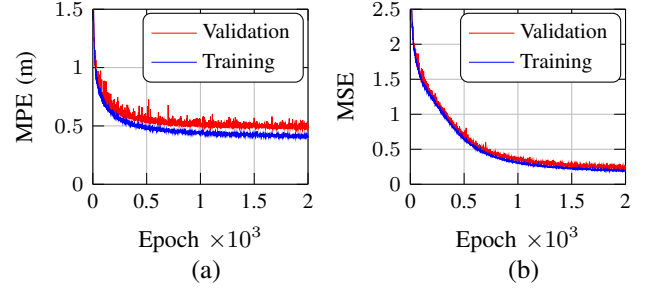


Fig. 3: MPE (a) and MSE (b) versus epochs for training and validation data considering 64 sensors.

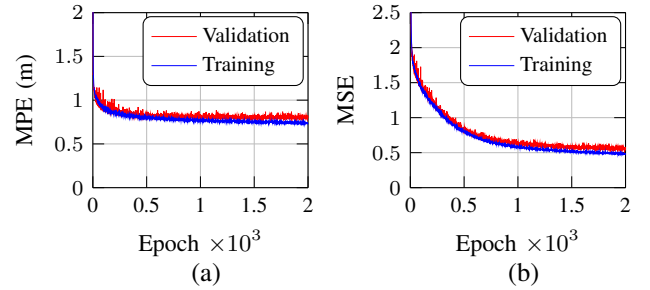


Fig. 4: MPE (a) and MSE (b) versus epochs for training and validation data considering 16 sensors.

DNN approach, the final estimator is obtained via indirect optimization, i.e., it is expected that by minimizing the MSE, the MPE will also be minimized. Therefore, measuring the training process via MPE ensures that the final weights and biases offer the best localization performance. Figure 3 shows the training and validation MPE and MSE versus training epochs for the scenario with 64 sensors, while Fig. 4 refers to the scenario with 16 sensors. From these figures, it is possible to conclude that the selected architecture and hyperparameters are providing adequate generalization performance, since the validation MPE/MSE values are closely following the training curves.

IV. PERFORMANCE EVALUATION

Considering a square room of size 20×20 meters, two scenarios with different sensor densities are assumed. In the first, 64 sensors are placed in a grid pattern with density of $6.25 \text{ m}^2/\text{sensor}$, and the second scenario has 16 sensors yielding a $25 \text{ m}^2/\text{sensor}$ density. Numerical simulations are performed to evaluate the accuracy of the presented localization schemes in terms of root mean squared error (RMSE) (m) using the model given by (1) subject to different shadowing variances. The path loss exponent is fixed to $\beta = 3.23$, which corresponds to an office like building (indoor) [8]. The transmit power is assumed fixed to 20 dBm, and the decorrelation distance $d_{\text{cor}} = 1 \text{ m}$. In addition, for providing insights about the largest localization errors, the empirical cumulative distribution function (CDF) of the localization

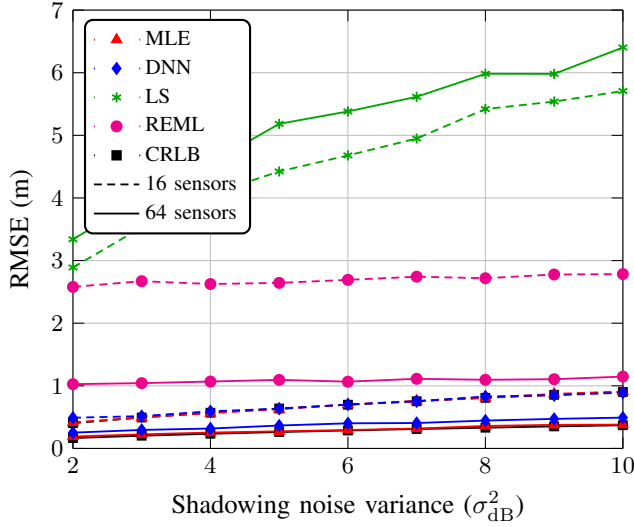


Fig. 5: CRLB and MPE for MLE, DNN and LS estimators for first (solid lines) and second (dashed lines) scenarios.

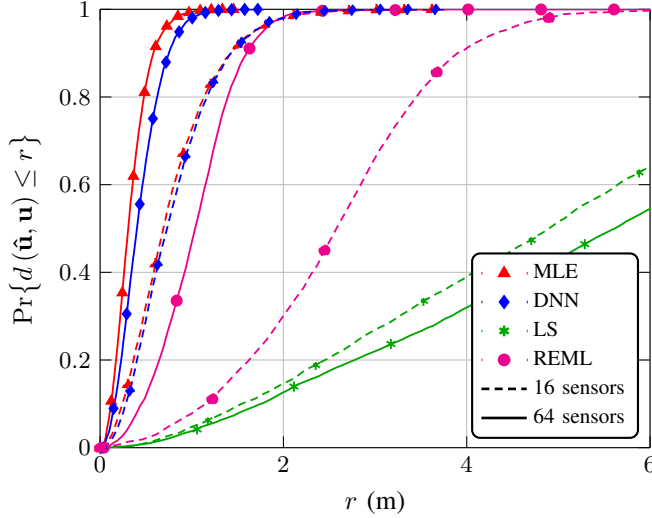


Fig. 6: Empirical CDF of localization error considering $\sigma_{dB}^2 = 10$ dB for first (solid lines) and second (dashed lines) scenarios.

error is also presented. For the DNN training, synthetic data have been generated using (1). The shadowing noise variance of the training set has been fixed to 10 dB, and the other parameters remain the same. Note that for training and testing, independent data sets were generated.

A. Localization Error Analysis

Figure 5 shows the localization performance of the different schemes for the two scenarios. In the first scenario, there are 64 sensors, and consequently, more information is available to the estimation algorithms. Therefore, the localization performance is improved when compared with the scenario where there are only 16 sensors in place. As expected, the localiza-

tion performance degrades as the shadowing noise variance increases. This is in accordance with the intuition that more random movement in the area leads to poorer localization performance. Interestingly, this observation is not valid for the LS estimator. This behavior can be explained with the fact that noisy measurements appear in the matrix $\mathbf{X}$, and the inversion in (7) leads to noise enhancement when N increases. The performance of REML does not change significantly with shadowing noise variance, since the localization inaccuracies mainly come from the assumption that the highest value within the REM corresponds to the transmitter position. Even though the MLE and DNN estimators are biased, the CRLB is used as a performance bound, since it can be easily derived for such system model, and provides insightful hints about the estimators' performance. Both MLE and DNN present sub one meter accuracy under both scenarios. This result shows that DNN is an attractive alternative to MLE.

Figure 6 shows the empirical CDF of the localization error assuming a fixed shadowing noise variance $\sigma_{dB}^2 = 10$. It can be seen that, DNN and MLE provide better localization accuracy than LS, presenting localization errors lower than 1 meter for 80% of the time for 64 sensors, and 65% with 16 sensors. This is expected since MLE holds asymptotic optimal performance and the DNN estimator can approximate the MLE. The LS has a poor performance since noisy measurements appear in the matrix $\mathbf{X}$, rather than in the observation vector $\mathbf{y}$.

B. Computational Complexity Analysis

The number of real multiplications of the online DNN estimator can be written w.r.t. the network architecture as

$$C_{DNN} = \sum_{l=1}^{L-1} (N_u^{(l-1)} + 1) N_u^{(l)}, \quad (15)$$

where $N_u^{(l)}$ represents the number of units in l -th layer of the network, L is the total number of layers including input and output layers.

The total number of real multiplications necessary for the LS estimator in (7) can be expressed as

$$C_{LS} = 36N + 64, \quad (16)$$

where N represents the total number of sensors.

The MLE has been implemented as a full search over the quantized area of interest with a minimum resolution of 10 cm, which corresponds to 40,000 discrete points over the assumed 20 by 20 meters area. Therefore, the required number of multiplications is given by

$$C_{MLE} = NA/q^2, \quad (17)$$

where q represents the grid resolution in meters and A denotes the area in meter².

REML also uses a pixel resolution of 10 cm, its complexity is given by [2]

$$C_{REML} = KN \log(N). \quad (18)$$

TABLE II: Complexity comparison among estimators

Scenario	Estimator	Number of real multiplications
16 sensors	DNN	35,458
	MLE	640,000
	LS	640
	REML	4,436,141
64 sensors	DNN	41,602
	MLE	2,560,000
	LS	2,368
	REML	2.661,685

Table II shows a complexity comparison among the transmitter location estimation algorithms. Under these scenarios, one can see that LS has the lowest computational complexity among the presented algorithms due to its linear formulation. However, DNN estimator is capable of achieving performance comparable to MLE without significant increase in complexity when compared to the LS solution. The complexity reduction achieved by the DNN estimator corresponds to a reduction of 98.37% for the first scenario, and 94.46% for the second when compared to the MLE. Even though REML has larger complexity, it holds an advantage over MLE, since it does not require knowledge on the transmit power and path loss exponent, which were assumed to be known in the presented analysis. It is important to point out that the complexity analysis shown here corresponds to the online phase. The offline training complexity has not been taken into account, since it should be performed only once for a given scenario. The analysis on the required frequency of network weight updates remains as further work on the scope of DNN based localization.

V. CONCLUSION

The blind transmitter localization problem has been tackled using a DNN based estimator. It is shown that DNN is capable of achieving performance in terms of localization error very close to the optimal MLE while keeping a reduced computational complexity. Furthermore, significant improvement when compared to the low-complexity LS can be achieved.

Extension to scenarios where multiple users are present is a natural extension of the work presented here. Additionally, aspects such as robustness against model imperfections, estimators' performance comparison when measured data sets are available, and frequency of weights update shall be investigated in future works. In essence, it has been shown that DNN based localization has the potential to be a key technique to provide accurate and reliable location information for improving services in future wireless communications systems.

APPENDIX

DERIVATION OF CRLB FOR TRANSMITTER LOCALIZATION

Assuming fixed and known path loss exponent and transmit power, the likelihood function of the measurement vector $\mathbf{m}$, under the model defined by (1), is given by

$$p(\mathbf{m}; \mathbf{u}) = \frac{1}{(2\pi)^{\frac{N}{2}} \det^{\frac{1}{2}}(\mathbf{C})} \times \exp\left(-\frac{1}{2}(\mathbf{m} - \boldsymbol{\mu}(\mathbf{u}))^T \mathbf{C}^{-1}(\mathbf{m} - \boldsymbol{\mu}(\mathbf{u}))\right), \quad (19)$$

where $\boldsymbol{\mu}(\mathbf{u}) = P_0 - 10\beta \log_{10}\left(\frac{d(\mathbf{u}, \mathbf{v}_j)}{d_0}\right)$. The corresponding entries of the Fisher information matrix are defined as [10]

$$[\mathbf{I}(\mathbf{u})]_{i,j} = -\mathbb{E}\left[\frac{\partial^2 \ln p(\mathbf{m}; \mathbf{u})}{\partial [\mathbf{u}]_i \partial [\mathbf{u}]_j}\right]. \quad (20)$$

The CRLB on the variance of the estimated transmitter coordinates is given by

$$\text{var}(\hat{\mathbf{u}}) \geq \text{tr}(\mathbf{I}(\mathbf{u})) \quad (21)$$

$$\geq \frac{[\mathbf{I}(\mathbf{u})]_{1,1} + [\mathbf{I}(\mathbf{u})]_{2,2}}{[\mathbf{I}(\mathbf{u})]_{1,1}[\mathbf{I}(\mathbf{u})]_{2,2} - [\mathbf{I}(\mathbf{u})]_{1,2}^2}, \quad (22)$$

where

$$[\mathbf{I}(\mathbf{u})]_{1,1} = \left(\frac{10\beta}{\ln(10)}\right)^2 \mathbf{1}^T \mathbf{C}^{-1} \frac{(u_x - \mathbf{v}_x)^2}{d^4(\mathbf{u}, \mathbf{v})}, \quad (23)$$

$$[\mathbf{I}(\mathbf{u})]_{2,2} = \left(\frac{10\beta}{\ln(10)}\right)^2 \mathbf{1}^T \mathbf{C}^{-1} \frac{(u_y - \mathbf{v}_y)^2}{d^4(\mathbf{u}, \mathbf{v})}, \quad (24)$$

$$[\mathbf{I}(\mathbf{u})]_{1,2} = \left(\frac{10\beta}{\ln(10)}\right)^2 \mathbf{1}^T \mathbf{C}^{-1} \frac{(u_x - \mathbf{v}_x)(u_y - \mathbf{v}_y)}{d^4(\mathbf{u}, \mathbf{v})}, \quad (25)$$

where $\mathbf{1}$ is a N -long all-ones vector.

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